

Introduction to the circular flow of income

The circular flow of income

The circular flow of income model shows how income, spending and output move around an economy. Figure 16.2 shows the simplest model of the circular flow of income. The model shows that output generates income which is then spent on the output. This flow explains why GDP can be measured by calculating the country's output, income and expenditure.

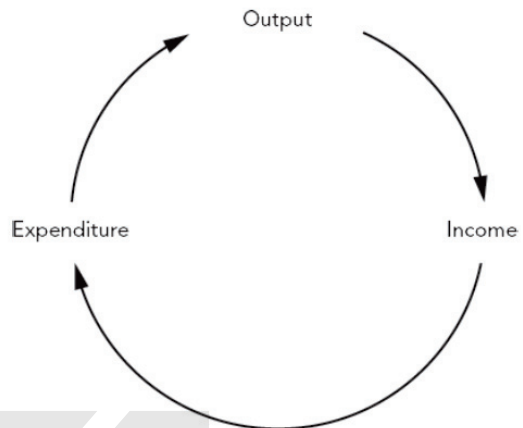


Figure 16.2: A simplified circular flow model

The difference between an open and a closed economy

An open economy is an economy that takes part in international trade. In contrast, a closed economy is one that does not export or import goods and services. Of course, no economy is a totally closed economy but such an economy can be used in a model of how an economy works.

The circular flow in a closed economy and an open economy

Figure 16.3 shows the process of circular flow in a closed economy. The inner circle shows the real flow of products and factor services and the outer circle the money flow of spending and incomes.

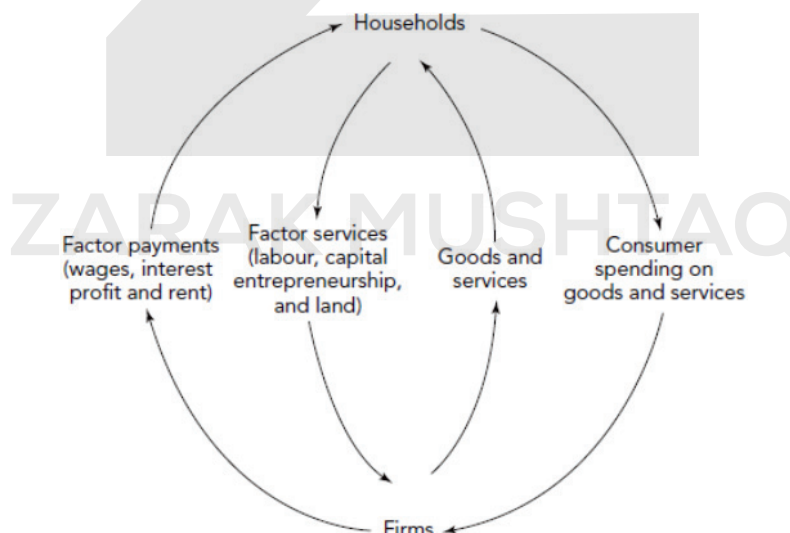
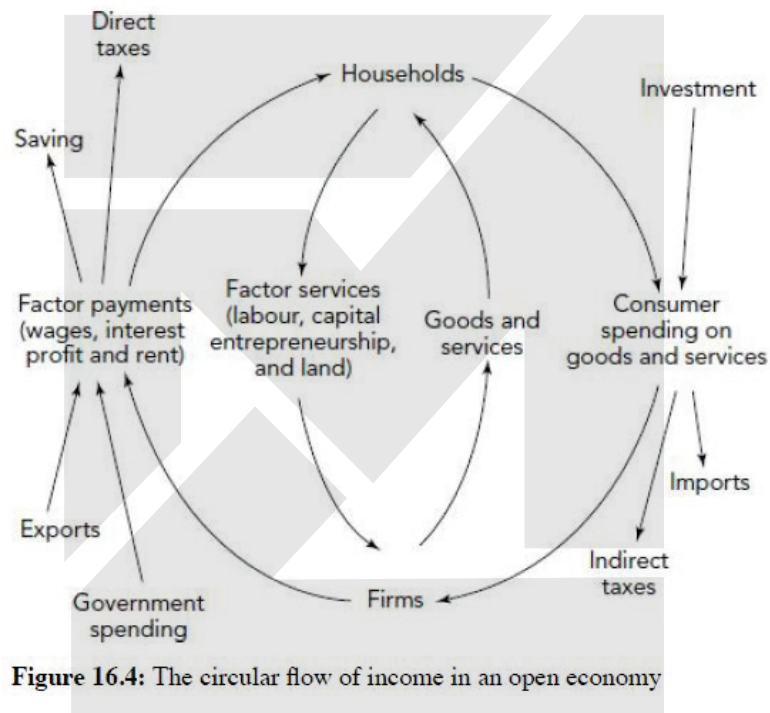


Figure 16.3: The circular flow of income in a closed economy

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The impact of injections and leakages on the circular flow

In practice, there are leakages out of the circular flow - some income is saved, some is taxed and some is spent on imports. Some expenditure is also additional to the spending that comes from the incomes generated by domestic output. These extra items of spending, known as injections, are investment, government spending and spending by foreigners on a country's exports. Figure 16.3 includes these leakages, also known as withdrawals, and the injections, in a circular flow of income diagram of an open economy.



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National income statistics

National income statistics

National income is a country's total output. People earn an income from producing the output. This income is then spent on the output. This means that total output should equal total income and total expenditure.

National income statistics is a general term for a number of measures of a country's economic activity in terms of its output, income and expenditure. A government measures a country's total output to assess the performance of the economy. An economy is usually considered to be doing well if its output is growing at a sustained and sustainable rate. A government is also likely to compare its country's output and the growth of that output with those of other countries. International organizations, including the United Nations, also make use of national income statistics to compare the performance of different countries. Higher output has the potential to increase people's living standards. If an economy is growing at a slower rate than it is considered capable of, a government is likely to introduce policy measures to stimulate the economy.

Economists examine both the level and rate of growth of a country's output. For instance, in 2019 the output of Malaysia increased to \$354 billion from \$319 billion the previous year and the output of the USA increased from \$19.52 trillion to \$20.58 trillion. The US experienced a greater absolute increase in output but a smaller percentage increase than Malaysia as it was starting from a larger base (\$19.52 trillion compared with \$319 billion).

Gross domestic product and gross national income

Gross domestic product

The most widely used measure of national income is known as gross domestic product (GDP). GDP is used by economists, governments and international organizations to assess what is produced, earned and spent in an economy. 'Gross' means total, 'domestic' refers to the home economy and product means 'output'. So, for example, Pakistan's GDP is a measure of the total value of output produced by the factors of production based in Pakistan in a year.

Gross national income

Gross national income (GNI) is increasing in importance as a measure. GNI is included, for instance, by the United Nations in its Human Development Index. GNI goes further than GDP in changing the focus from output produced in a country to income earned by the country's residents and firms regardless of where it is earned. As well as adding net property income from abroad, GNI also includes other sources of income that residents receive from abroad and deducts other sources of income that foreigners receive from

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the country. These payments are net receipts of compensation of employees and net taxes less subsidies on products.

Compensation of employees is wages earned by workers who are resident in one country but who work abroad for short periods. These include seasonal and cross-border workers.

Some tax revenue on products may be paid to other countries and international organizations and some production subsidies may be received by other governments and international organizations. For example, Germany pays some of its tax revenue to the European Union and also receives some production subsidies from the European Union. Governments also measure gross national disposable income. This includes income sent home to relatives by people working abroad minus income sent by foreigners working in the country to their relatives abroad.

Differences in countries' GDP and GNI

Most countries' GDP and GNI are relatively similar. However, in the case of some countries, there is a clear difference between their GDP and GNI. A number of countries, including Ireland, have a noticeably higher GDP than GNI. This is because foreign multinational companies (MNCs) and foreign workers make an important contribution to the output of these countries. A major reason for the rapid rise in Ireland's GDP in recent years is its ability to attract foreign investment. However, foreign investment has also resulted in a net outflow of profits and other income.

Other countries have a higher GNI than GDP because they receive a net inflow of property income from abroad. German firms, for example, have invested large amounts in other countries in the past and now receive an inflow of profits from their multinational companies operating in other countries. A number of countries receive a significant inflow of income from their citizens working abroad, for example Nepal.

Methods of measuring GDP

There are three ways of calculating GDP: the output, income and expenditure methods. Each method should give the same total because they all measure the circular flow of income produced in an economy. So the value of output is equal to the incomes that are earned producing it, meaning wages, rent, profit and interest. If it is assumed that all incomes are spent, expenditure will, by definition, equal income.

The output method

The output method measures the value of output produced by industries such as manufacturing, construction, distributive, hotel and catering and agricultural industries or sectors. It is also sometimes called the production method.

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When using the output measure it is important to avoid counting the same output twice. For example, if the value of cars sold by manufacturers is added to the value of output of the firms which make the tyres, double counting will occur. To avoid double counting, output is measured either by totalling the value of the final goods and services produced or by adding the value added at each stage of production. Value added is the difference between the price a firm pays for the goods and services it buys from other firms and the price it sells its product for. For example, a TV manufacturing firm buys components costing \$280 000 and uses them to make TVs that it sells for \$350 000. The firm has added \$70 000 to output. The \$280 000 has already been counted and then the \$70 000 is added. Adding the value at each stage of production should give the figure as the market value of the finished product produced.

The income method

The value of an output produced is based on the costs involved in producing that output. The costs include wages, rent, interest and profits. All these payments represent income paid to factors of production. For instance, workers receive wages and entrepreneurs receive profits. In using the income method it is important to include only payments received in return for providing a good or service.

So transfer payments, which are transfers of income from taxpayers to groups of individuals for welfare payments and to firms in the form of subsidies, are not included.

The expenditure method

Using the expenditure method, GDP is calculated by adding up consumer expenditure (spending by households), government spending on goods and services, total investment, changes in stocks and the difference between exports and imports. As with the income method, transfer payments are not included. In this case, government spending on welfare payments does not represent spending on goods and services. It is spending which transfers income to the country's households and firms.

What is produced in a year will either be sold or added to stocks. So, if additions to stocks are added to expenditure on goods and services, a measure is obtained that will equal output and income. In using this method, it is necessary to add expenditure on exports and deduct expenditure on imports. This is because the sale of exports represents the country's output and creates income in the country, whereas expenditure on imports is spending on goods and services made in foreign countries and creates income for people in those countries.

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Market prices and basic prices

GDP and GNI are measured both in market prices and basic prices. Market prices are the prices charged to consumers. They include any taxes on products (indirect taxes) that have been imposed and they deduct any subsidies that have been given to producers.

Basic prices (also referred to as factor cost) are the prices which would be charged without government intervention and which equal the income paid to the factors of production for making the output. To get from market prices to basic prices, taxes on products are deducted and subsidies on products are added. For example, if a product which sells for \$12 has a \$4 tax on it, the market price would be \$12 but the basic price would be \$8. A product that a producer would sell for \$15 but which is subsidized at \$2 per unit, will have a basic price of \$15 but a market price of \$13.

Gross values and net values

GDP and GNI include gross investment that is total investment. Gross investment includes the output of capital goods both used to replace existing capital goods that have worn out or become out of date due to advances in technology and capital goods required to expand capacity. Net domestic product (NDP) and net national income (NNI) only include net investment. They deduct the value of the replacement capital goods. This value is referred to as depreciation or capital consumption.

Net investment gives an indication whether the country's ability to produce goods and services in the future will increase, stay the same or even decrease.

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